Stage 1: Niti Vahan

> Welcome, explorers, to **Niti Vahan**, one of the thrilling themes of eYRC 2026-27! This year you build an autonomous vehicle that treats the rules of the road not as an obstacle, but as its own logic.

What This World Is About

The Disciplined City

Picture a city arena where an Ackermann-steered vehicle has to earn its way through traffic reading lanes, holding at junctions, signalling its turns, and never crowding a neighbour's space. Beyond just following rules, the Vahan is defined by its deep civic sense: because it constantly predicts its own future path, it never makes the sudden, selfish moves that cause others to panic.

The Mission

The core task is pick-up and drop-off: collecting a passenger and threading it through the arena while obeying every traffic rule along the way. Stage 1 is about proving this capability in simulation.

The Challenge

Your mission is to bring this disciplined driver to life in simulation first, then on real hardware:

- Encode the traffic rulebook as mathematical constraints that a predictive control

system can plan against.

- Generate a disciplined path that stops at junctions, signals turns, and completes milestones with minimal rule violations.
- Maintain dynamic stability while respecting the vehicle's physical limits when navigating between locations.
- Execute precise parking within a designated zone to complete the route.

It's about proving that smart control logic can produce safer, more orderly movement through the city.

The Abilities You Will Build

Domain	Skills
Perception	Camera-based lane detection, junction and traffic-signal recognition
Planning & Control	Predictive path planning under traffic-rule constraints, Ackermann-steering dynamics, stable path tracking
Decision-Making	A layer that governs when to proceed, hold, or divert toward the parking zone

The Two Stages

ⓘ Stage 1 - Simulation Challenge

Build and validate lane detection, lane switching via obstacle avoidance, sign detection, and traffic-rule compliance.

🚀 Stage 2 - Hardware Build

Realize the same pick-up and drop-off task on a physical platform.

Win Condition

A robot that can navigate the arena by following lanes, avoiding collision, and parking in the designated area.

This is your starting point. Everything you build across both stages - the lane keeping, the junction discipline, the parking - traces back to the mission laid out here.